Case Study

Organization: Silver Oak University, Ahmedabad, Gujarat

Department: MSc. (Data Science)

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Title: Information Security Identification of Information Assets, Threats, Vulnerabilities and Risks Assessment

1. Introduction:

Silver Oak University is a private university located in Ahmedabad, Gujarat, India. The university was recognised by the University Grants Commission (UGC) in 2019 after being declared a State Private University under the Gujarat Private Universities Act, 2009.

The university offers diploma, undergraduate, postgraduate and doctoral programmes across engineering, technology, management, commerce, design, law, science, humanities, healthcare and allied disciplines. More than 25,000 students study across its constituent colleges and institutes.

In May 2025, Silver Oak University received Grade A accreditation from the National Assessment and Accreditation Council (NAAC). The university describes itself as the youngest university in Ahmedabad to achieve this distinction.

2. Information Assets:

An information asset is any data, system, application, device, or information resource that has value to the university.

No.	Information Asset	Examples of Data	Importance
1	Student Database	Name, DOB, address, contact details, enrolment data	Very High
2	Academic Records	Marks, grades, attendance, certificates	Very High
3	Examination Data	Question papers, answer sheets, results	Critical
4	Financial Data	Fees, transactions, bank/payment information	Very High
5	Faculty Data	Employee details, salary, qualifications	High
6	HR Data	Employee records, documents, payroll	Very High
7	Login Credentials	Student/faculty/admin usernames and passwords	Critical
8	Research Data	Research papers, datasets, intellectual property	High
9	University Website	University information and announcements	Medium
10	Email System	Official communications and attachments	High
11	ERP System	Administration, finance, academics	Critical
12	Backup Data	Copies of databases and important files	Critical
13	Network Infrastructure	Routers, switches, firewalls, servers	Critical
14	CCTV/Access Data	Video footage, entry/exit information	High

3. Threats:

A Threat is something capable of causing harm to an information asset.

Major threats to a Silver Oak University

External Threats	Internal Threats
Phishing attacks	Disgruntled employee
Ransomware	Unauthorized access by staff
Malware	Student misuse of accounts
Hacking	Accidental deletion of data
Password attacks	Sharing passwords
Website defacement	Unauthorized copying of confidential data
SQL injection	Misconfiguration by IT staff

4. Vulnerabilities:

A vulnerability is a weakness that can be exploited by a threat.

Vulnerability	Explanation
Weak passwords	Users may use simple passwords
No MFA	Accounts may be compromised using stolen passwords
Outdated software	Unpatched systems may contain security vulnerabilities
Poor access control	Users may have access to data they don't require
Unencrypted data	Stolen data can be read easily
Poor backup strategy	Data may not be recoverable after ransomware

Insecure Wi-Fi	Attackers may gain unauthorized network access
Phishing awareness gap	Users may click malicious links
SQL injection vulnerabilities	Web applications may allow database manipulation
Unsecured endpoints	Student/faculty computers can introduce malware

5. Threat–Vulnerability–Risk Assessment:

Information Asset	Threat	Vulnerability	Risk	Impact
Student Database	Data theft	Weak access controls	Unauthorized disclosure of student information	Very High
Examination Database	Insider attack	Excessive privileges	Exam papers/results may be leaked or manipulated	Critical
ERP System	Ransomware	Unpatched server	University operations may stop	Critical
Student Portal	Hacking	SQL injection	Student records could be modified	Very High
Faculty Accounts	Phishing	Lack of MFA	Attacker may obtain	High

			account access	
Research Database	Data theft	Weak authentication	Intellectual property may be stolen	High
University Website	Website attack	Outdated CMS	Website may be defaced	Medium
Wi-Fi Network	Unauthorized access	Weak Wi-Fi security	Attacker may enter university network	High
CCTV System	Unauthorized access	Default credentials	Surveillance footage exposed	High
Email System	Phishing/malware	Poor awareness	Credentials or sensitive documents stolen	High

6. Detailed Risk Example:

Scenario: Student Database Breach

- **Assets:** Student Database which contains Student name, Enrolment number, Contact information, Date of birth, Address, Academic records, Attendance, Course information, Documents
- **Threat:** A cybercriminal attempts to steal student information.
- **Vulnerability:** Weak passwords, No multi-factor authentication, Excessive user permissions, Poor database monitoring
- **Risk:** The attacker obtains unauthorized access to the student database and downloads student information.

7. Recommended Security Controls:

The university can reduce these risks by implementing appropriate controls.

A. Access Control

- Role-Based Access Control (RBAC)
- Strong password policies
- Multi-Factor Authentication
- Regular access reviews
- Disable accounts of former employees

B. Data Security

- Encryption during transmission
- Data classification
- Secure data deletion
- Restrict USB/storage access where appropriate

C. Network Security

- Firewall
- Secure Wi-Fi
- VPN for remote administration
- Separate student, faculty and administrative networks

D. Application Security

- Regular vulnerability assessments
- Secure coding practices
- SQL injection prevention
- Web Application Firewall

- Penetration testing

E. Backup & Recovery

- 3 copies of important data
- 2 different storage media
- 1 copy offline/off-site

F. Employee & Student Awareness Training on

- Phishing
- Password security
- Malware
- Data privacy
- Safe internet usage
- Reporting suspicious activities

8. CIA Triad:

- **Confidentiality:** Only authorized people should access university data.

  Example: Students should not be able to access another student's marks.

- **Integrity:** Data must remain accurate and unchanged by unauthorized users.

  Example: A student should not be able to change their examination marks.

- **Availability:** Authorized users should be able to access information when required.

  Example: The examination system should be available during result publication.

9. Conclusion:

A Silver Oak University possesses a large number of valuable information assets, including student records, examination information, financial data, employee information, research data, credentials, and institutional systems.

The major threats include **phishing, ransomware, malware, unauthorized access, insider threats, data theft, and application attacks**. These threats become more dangerous when vulnerabilities such as weak passwords, lack of MFA, outdated software, excessive privileges, inadequate backups, and poor security awareness exist.

Therefore, the university should adopt a **risk-based information security approach** combining technical, administrative, and physical controls. Regular risk assessments, access reviews, security awareness programs, vulnerability assessments, backups, encryption, monitoring, and incident-response planning can significantly reduce the university's overall cybersecurity risk.